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POSTER SESSION

Cost & Energy of Everyday LLM Workloads: A Visual Field Guide for Python Developers

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RESULTS AT A GLANCE

Bigger ≠ Better
Scaling model size often raises energy, latency, and VRAM faster than toy-task quality.

Quantization Changes Economics
q4/q8 variants can cut memory and energy enough to change where models can run.

No Single Winner
The best model depends on workload: code, summary, chat, RAG, or embeddings.

Hardware Depends on Scale
CPU can be fine for tiny local jobs; GPU wins when batching or concurrency grows.

OVERVIEW

Everyday LLM calls look simple from Python, but the real deployment decision is a tradeoff between **quality**, **latency**, **VRAM**, **energy**, and **cost**.

We benchmark chat completions, document summarization, classification-style prompts, RAG-style search, code generation, and batch embeddings across model sizes, quantization levels, architectures, and CPU/GPU configurations.

Thesis: the most accurate LLM is often not the most economically efficient one. LLM deployment is a systems engineering problem, not just model selection.

BENCHMARK SCOPE

19	5	234	L40S
models / variants	workload families	metric rows	NVIDIA L40S

Question	Measured Signals	Why It Matters
Bigger model?	toy score, latency, GPU joules, VRAM	find diminishing returns
Quantize?	q4/q8/fp16, GPU tokens/J	fit smaller hardware
Which workload?	latency slices, GPU joules, cost, score	avoid one-model thinking

MEASUREMENT PIPELINE

Each run produces one row of metrics: latency slices, token counts, GPU active energy, memory, workload score, and request-cost estimates.

FIXTURES	Repeatable prompts and document sets for each workload.
CLIENT	Ollama/OpenAI-compatible adapter wraps model calls and response logs.
ENERGY	NVML samples GPU power/utilization/VRAM at 0.1s (~10Hz). CPU package joules require RAPL/bare metal; this VM only has a labeled TDP estimate.
VISUALS	pandas + matplotlib generate latency waterfalls, cost bars, and active-energy figures.

Energy attribution matters: measured GPU active energy is used for model-energy claims. CPU TDP/utilization estimates are retained in the CSV but excluded from measured-energy charts.

WHAT GETS TIMED

Slice	Meaning	Why Python Devs Care
Network	client call overhead	endpoint placement, batching
Tokenization	prompt processing / prefill	long docs are not free
Inference	model generation	model size dominates
Post-processing	parse, serialize, rank	pipeline glue is measurable

MEASURED GPU ENERGY TRACE FOR ONE LLM

Measured example: CodeLlama 7B Q4 chat on L40S — CPU + RAM + GPU + VRAM



How to read it: one CodeLlama 7B Q4 chat request on L40S. The cards show measured idle-adjusted GPU joules, raw GPU joules, latency, and peak VRAM. CPU package joules were unavailable in the VM and are not used as measured energy.

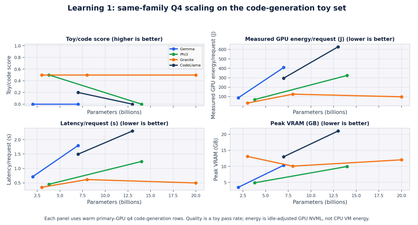
LEARNING 1: BIGGER MODELS HAVE DIMINISHING RETURNS

SAME ARCHITECTURE · DIFFERENT SIZES

Scaling parameters often increases infrastructure cost faster than model quality.

Same-family comparisons isolate parameter count: Gemma 2B/7B, Phi3 3.8B/14B, Granite 3B/8B/20B, CodeLlama 7B/13B where present.

Steady-state measured benchmark: toy/code score, GPU energy, latency, and VRAM — SAME-FAMILY SCALING



Each panel uses warm primary-GPU q4 code-generation rows. Quality is a toy pass rate; energy is idle-adjusted GPU NVML, not CPU VM energy.

Score note: the quality panel is a tiny code-generation pass rate, not a claim that lower-parameter models are universally smarter. Cold model-load rows are excluded from this comparison so the plot reflects steady-state requests.

LEARNING 2: QUANTIZATION CHANGES THE ECONOMICS OF AI

SAME MODEL · DIFFERENT PRECISION

4-bit and 8-bit models often retain useful quality while reducing memory, energy, and latency.

Measured benchmark: fp16 vs q8 vs q4 only within the same model — % OF SAME-MODEL FP16 BASELINE

Takeaway: q4 averaged 48% VRAM, 47% energy, 51% latency, and 2.9x throughput versus FP16; toy-score retention was 100%.

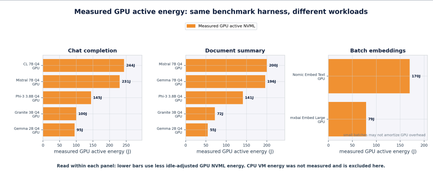
How to read it: FP16 is the 100% baseline for the exact same model; q8/q4 are compressed representations. Left: costs to minimize (VRAM, energy, latency). Right: benefits to preserve or improve (throughput, toy-score retention). Bars average only complete same-model FP16/q8/q4 triplets.

LEARNING 3: ENERGY DEPENDS ON THE WORKLOAD

COMMON WORKLOADS · MEASURED GPU JOULES

Chat, document summaries, RAG, and embeddings do not stress the same parts of the stack.

Measured benchmark: GPU active energy by scenario — JOULES / REQUEST · LOWER IS BETTER



How to read it: each panel is a different workload. These are measured idle-adjusted GPU joules, not CPU energy. A model/hardware pair that is cheap for chat may not be cheap for long summaries or embedding batches.

CPU VS GPU: THE SCALE INFLECTION POINT

HARDWARE SCALE · CONCEPTUAL HEURISTIC

Hardware choice changes when prompts, outputs, batching, or concurrency grow. Single-request latency is not throughput, and CPU energy was not measured in this VM.

Conceptual deployment heuristic, not measured concurrency data — CPU-ONLY VS GPU + BATCHING

CPU is reasonable for
- prototyping
- local scripts
- tiny embedding jobs
- single-user offline tools
- low request volume

GPU becomes better for
- long prompts / outputs
- batch embeddings
- multi-user chat
- latency-sensitive services
- high concurrency

Tiny embedding jobs may not amortize GPU overhead. Measure batch size before assuming the GPU path wins.

PRACTICAL HEURISTICS

#	Heuristic
1	Benchmark the workload, not the model card.
2	If quality barely improves but joules rise, use the smaller model.
3	If VRAM is the deployment constraint, use q4/q8.
4	If the workload changes, re-benchmark; do not reuse one winner.
5	If concurrency grows, use GPU batching.
6	If the workload is tiny and local, CPU may be good enough.
7	Track measured GPU tokens/joule; use bare-metal RAPL or a wall meter for CPU energy.